

Question ID: 571174f3

| Assessment | Test | Domain  | Skill                                            | Difficulty                                        |
|------------|------|---------|--------------------------------------------------|---------------------------------------------------|
| SAT        | Math | Algebra | Systems of two linear equations in two variables | Hard <div><div></div><div></div><div></div></div> |

Question

$$\frac{2}{5}x + \frac{7}{5}y = \frac{2}{7}$$
$$gx + ky = \frac{5}{2}$$

In the given system of equations,  $g$  and  $k$  are constants. The system has infinitely many solutions. What is the value of  $\frac{g}{k}$ ?

Correct Answer: .2857, 2/7

Rationale

The correct answer is  $\frac{2}{7}$ . It's given that the system has infinitely many solutions. A system of two linear equations has infinitely many solutions if and only if the two linear equations are equivalent. Multiplying each side of the first equation in the system by  $\frac{35}{4}$  yields  $\frac{35}{4}(\frac{2}{5}x + \frac{7}{5}y) = \frac{35}{4}(\frac{2}{7})$ , or  $\frac{7}{2}x + \frac{49}{4}y = \frac{5}{2}$ . Since this equation is equivalent to the second equation and has the same right side as the second equation, the coefficients of  $x$  and  $y$ , respectively, should also be the same. It follows that  $g = \frac{7}{2}$  and  $k = \frac{49}{4}$ . Therefore, the value of  $\frac{g}{k}$  is  $\frac{\frac{7}{2}}{\frac{49}{4}}$ , or  $\frac{2}{7}$ . Note that 2/7, .2857, 0.285, and 0.286 are examples of ways to enter a correct answer.